

Library Management System

Functional Requirements

System need to meet these requirements to function well.

Admin Functions

Admin is a user that can borrow a book with some more functions and privileges

add_book: a function for admin to add a book by passing the book title and author.

The admin can add a book.

The admin can add more than one copy of the book.

The book may be existed, if the book already exists then the system should increment the count for the book copies.

The system should not add new record for the book (avoid duplication).

delete_book_record : a function for admin to delete the book completely by passing the book title and author.

The admin can remove the book from the system.

The book may not be existed. then print ("the book is not existed")

decrement_copies: a function for admin to remove one copy from a specific book by passing the book title and author.

Admin may want to drop one copy of the book because of some damage.

The number of copies available then have to be more than 1.

display_all_books: a function to display all books in the library. All books can be displayed by printing the books_df.

display_all_users: a function to display all users in the library. All users can be displayed by printing the users_df.

display_all_borrowed_books: a function to display borrowed books. All borrowed books can be displayed by printing the borrowed_books_df.

display_all_transactions: a function to display transactions (borrowing and returning). All transactions can be displayed by printing transaction_log_df.

*Note: for better practice books_df, users_df , reservations_df , borrowed_books_df , transaction_log_df , havetoreturn_df All these dataframes should be private only to admins but they are not in this project.

User Functions:

user created by passing the First Name, Last Name, age , gender

_update_global_overdue_books: a function that is automatically called during functions like borrow_book, return_book to update overdue books and apply fines.

Check books that are overdue from the user borrowed books.

Calculate fine by checking the current date and the return date that the user should return it in.

Add the user record and the book the user borrowed in the global havetoreturn_df dataframe that include the users that are late to return the book and the fines required from the user.

borrow_book : a function for user to borrow a book by passing book title and author name and how many days he/she want to borrow the book.

The user wants to borrow a book.

The book needs to be in the library and currently available that means that available copies >1 so we need to check.

The book when borrowed need to be add to the local dataframe borrowed_books dataframe.

The book when borrowed need to be add to the global dataframe borrowed_books_df.

We need to log the transaction for borrowing the book in transaction_log_df.

return_book: a function for user to borrow a book by passing book title and author name.

We need to check if the book is really borrowed by the user.

If the book is borrowed then we need to drop the book from the global and local borrowing book dataframes borrowed_books_df, self.borrowed_books.

We need to log the returning transaction in the transaction_log_df.

When the user returns the book there are other users may have reserved the book and they are currently waiting in a queue.

The first person on the queue should receive a notification.

We need to update the global overdue books after returning.

display_my_overdue_books: a function that enable the user to display books he/she is late to return.

view_borrowed_books: a function that enable the user to display books he/she borrowing now.

reserve_book: a function to enable user to reserve a book if not available yet by passing the book title and author.

If the book exists and all the copies are borrowed then users can reserve the book.

The reservation date will be the current date.

Add the reservation to the user reserved_books df.

Add the reservation to the global reservations DataFrame.

To handle multiple users reserving the book. The users will be added to a queue.

The first on to reserve is at the head of the queue. He/She will be the one to get notified when a user returns the book they want to borrow.

The function update_queue_positions will be called to sort reservations.

#note this transaction must be logged but it is not in this project.

cancel_reservation: a function to enable user to cancel reservation if no longer wanting the book by passing the book title and author.

To cancel reservation, we need to check if the reservation really exists.

Remove the user from the queue.

Remove the reservation from the user's reserved_books DataFrame.

Update the queue positions for other users.

Check if the book is currently available.

If the book is available, notify the first user in the queue.

display_user_details: a function to display user details.